



ME 207: Fluid Dynamics

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Lab Experiment 6: Reynolds Apparatus

Date of Experiment: - 29/3/2025

Date of Submission: - 2/4/2025

Group 14

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Abstract

In this experiment, we aim to characterize the behavior of the flow in relation to the Reynolds number and find the critical Reynolds number at which the flow transitions from the laminar to the turbulent.

1 Task 6.1

Q1: How is the Reynolds number (Re) defined for a flow? How is Re defined for flow through a circular tube?

The Reynolds number (Re) is a dimensionless quantity used to predict the flow regime in a fluid system. It is defined as the ratio of inertial forces to viscous forces in the flow:

The Reynolds number is defined as:

$$Re = \frac{uL}{\nu} = \frac{\rho uL}{\mu}$$

Where:

- ρ is the density of the fluid (SI units: kg/m³),
- u is the flow speed (m/s),
- L is a characteristic length (m),
- μ is the dynamic viscosity of the fluid (Pa·s or N·s/m² or kg/(m·s)),
- ν is the kinematic viscosity of the fluid (m²/s)

Reynolds number for the flow through the circular pipe.

For flow inside a circular pipe (or tube), the characteristic length is taken as the hydraulic diameter, which for a circular pipe is simply its diameter D . The Reynolds number is given by:

$$Re = \frac{\rho uD}{\mu} = \frac{uD}{\nu}$$

where D is the diameter of the pipe (m).

Flow Regimes in a Circular Tube Based on Re:

- **Laminar flow:** $Re < 2,300$
- **Transitional flow:** $2,300 \leq Re \leq 4,000$ (unstable, fluctuating)
- **Turbulent flow:** $Re > 4,000$

Q2: What is the physical significance of the Reynolds number? Show that it signifies the ratio between inertial and viscous forces in a fluid flow.

Engineering applications: The Reynolds number is essential in designing pipes, aircraft,

and other systems where fluid flow occurs. It helps predict how fluids behave in these systems, influencing factors like pressure drops, heat transfer, and drag forces.

Predicting mixing behavior: In industrial processes, such as mixing or chemical reactors, knowing the Reynolds number allows engineers to understand and control the rate of mixing and reaction efficiency.

Inertial forces

In a fluid flow, inertial forces are the forces that arise due to the motion of the fluid and its mass. These forces are proportional to the mass of the fluid and the acceleration it undergoes. The inertial force can be written as:

$$F_{inertial} = \rho v^2 L$$

where,

- ρ is the fluid density,
- v is the velocity of the fluid,
- L is the characteristic length (e.g., the diameter of a pipe or the size of an object in the flow).

This represents the force due to the fluid's motion over a characteristic length.

Viscous Forces:

Viscous forces, on the other hand, arise due to the internal friction between fluid layers moving at different velocities. The viscous force can be expressed as:

$$F_{viscous} = \mu \frac{v}{L}$$

where,

- μ is the dynamic viscosity of the fluid,
- v is the velocity of the fluid,
- L is the characteristic length.

This represents the frictional force that resists the flow, which is related to the fluid's viscosity and the velocity gradient.

The Reynolds Number:

The Reynolds number (Re) is defined as the ratio of inertial forces to viscous forces:

$$Re = \frac{F_{inertial}}{F_{viscous}}$$

Substituting the expressions for inertial and viscous force

$$Re = \frac{\rho v^2 L}{\mu \frac{v}{L}}$$

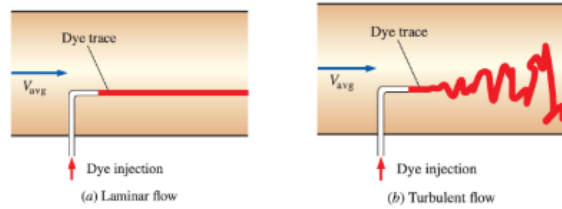
Which turns out to be

$$Re = \frac{\rho v L}{\mu}$$

Since the Kinematic viscosity is given as $\nu = \mu/\rho$, we can also represent Re as

$$Re = \frac{v L}{\nu}$$

Q3: How do you define a laminar and a turbulent flow? What causes a laminar flow transition into a turbulent one?



The flow regime in the first figure is called a laminar flow and is characterised by smooth streamlines and highly ordered motion, and the flow in the second figure is known as a turbulent flow, which is characterised by velocity fluctuations and highly disordered motions. The transition from laminar to turbulent flow does not occur suddenly. Instead, it happens over a region where the flow fluctuates between laminar and turbulent flows before it becomes fully turbulent.

Reynold's number is a non-dimensional number used to quantify a laminar flow and a turbulent flow; it is defined by the formula:-

$$Re = \frac{\rho v L}{\mu}$$

- ρ is the fluid density,
- v is the flow velocity,
- L is the characteristic length (e.g., pipe diameter),
- μ is the dynamic viscosity.

For the flow in the pipe,

- $Re < 2300$ is a laminar flow
- $2300 \leq Re \leq 4000$ is the transition flow
- $Re > 4000$ Turbulent flow

There are several reasons for a flow being transitioned into a turbulent flow from a laminar flow. Some of the reasons are as follows:

1. Sudden contractions or expansions in a pipe can create vortices in the flow, making the flow highly turbulent.
2. Small disturbances such as vibrations and noise can also make the flow turbulent.
3. Surface roughness in a pipe flow introduces disturbance in the boundary layer, which can disturb the flow and cause it to transition into a turbulent one.
4. Large temperature differences can also cause Turbulent flows (Ex:- Turbulence experienced in an aircraft).

Q4: What is the critical Re (Re_{cr}) for flow through a pipe, when the transition takes place? Do all flows have the same Re_{cr} at transition? If so, why? If not, report the Re_{cr} for various types of flows (such as flow over flat plates, flows around cylinders, etc.).

The Reynolds number at which the flow becomes turbulent is called the critical Reynolds number Re_{cr} . For internal flow in a circular pipe, the generally accepted value of the critical Reynolds number is $Re_{cr} = 2300$. The transition occurs after the Reynolds number increases beyond the critical value, which is $Re_{cr} \gtrsim 2300$, and it remains in the transitional flow for $2300 < Re < 4000$.

The value of the critical Reynolds number differs for different flow conditions as it depends on geometry, surface roughness and disturbance levels.

1. For flow over flat plates, the critical value of the Reynolds number is 500000.
2. For flow over an airfoil, the critical value of the Reynolds number is between 500000 - 3000000.
3. For flow around a circular cylinder, the critical value of Reynolds number is between 200000 - 300000.

2 Introduction

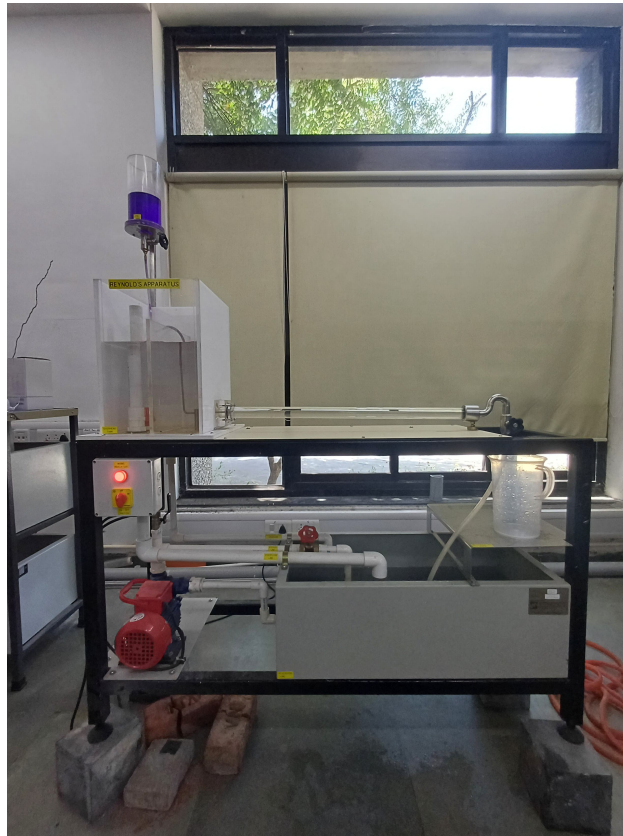


Figure 1: Image of the experimental setup.

The character of the flow of the fluid is of critical significance for the majority of the fields of engineering, like aerodynamics, pipe transport, and biological systems. The character of the flow is divided into two: turbulent flow and laminar flow. The fluid particles move in smooth or regular layers with minimal mixing in laminar flow, and in the turbulent flow, the speed of the fluid particle at any point changes continuously. The type of flow is determined by the Reynolds number (Re). Apart from the flow type, the Reynolds number has many practical applications like dimensional analysis and similitude.

In this experiment, the dye was injected into a laminar flow. While doing this, it forms a clear, well-defined line. It will mix with water minimally. The dye will rapidly mix with the water when the flow is turbulent.

First, we'll find the volumetric flow rate of the fluid using the tank. From this flow rate, the velocity of the fluid can be determined. Also, the Reynolds number of the flow will be determined from the fluid's velocity. Thus, the experiment is designed to obtain the Reynolds number of the flow, from which we can identify the flow type. The experiment also includes the calculation of uncertainties and experimental error when measuring the flow rates during the experiment.

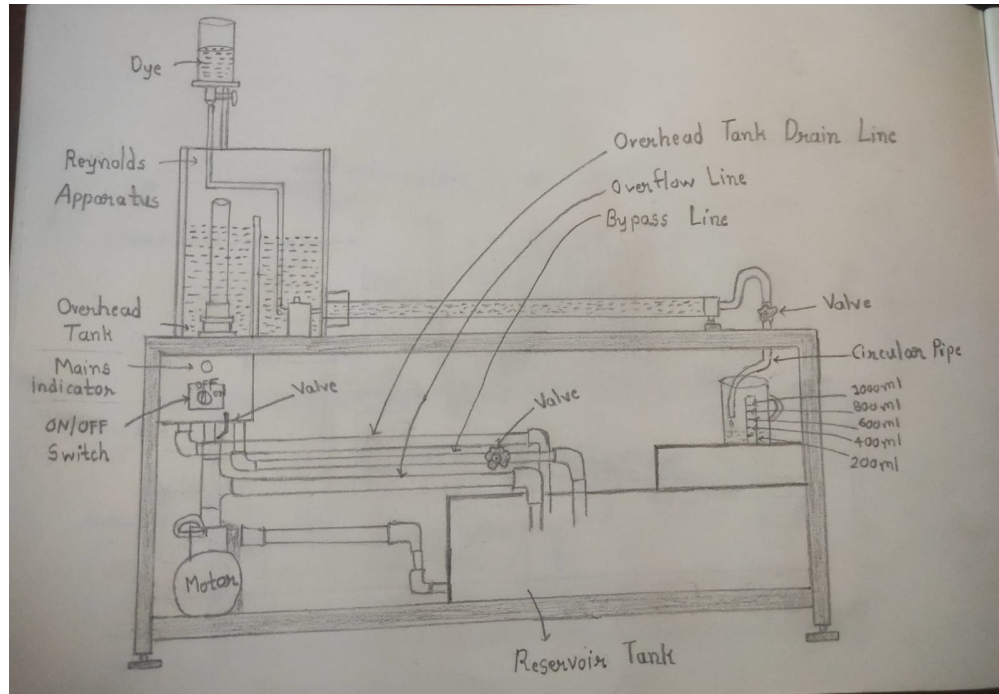


Figure 2: Schematic diagram

3 Objectives

- (i) Observe transition between laminar and turbulent flows in a circular tube.
- (ii) Experimentally estimate the critical Reynolds number prior to transition.
- (iii) Observe the behavior of a passive tracer in laminar and turbulent flow regimes

4 Experimental Procedure

1. Before starting the experiment, it is essential to check that the reservoir tank (sump tank) is filled with sufficient water.
2. Using the motor(pump), First switch on the motor. We filled the overhead tank by transferring the water from the reservoir tank to the overhead tank. After that, we switch off the motor to measure the accurate readings.
3. Starting the experiment with the flow in the tube at its minimum. The valve can control the flow rate as it is present on the right side of the tube, as shown in the image. Initially, We start with the minimum flow rate.
4. After setting the flow rate to its minimum value, The time for the water level to rise by 100ml in the small tub (which is with the measurement scale) is measured, and we can calculate the flow rate using the known values of the small tub.

5. Add the die to the flow by opening the valve near the die container. Then, we observed the die particles to check that they followed more or less in a straight line in the tube
6. We capture photos of die particles for each flow rate to check whether the flow is laminar or turbulent and when it will transition from laminar to turbulent.
7. We must repeat this process for the rest of the flow rates.
8. If the overhead tank is empty in the middle of the procedure, use the motor(pump); first, switch on the motor. We can fill the overhead tank by transferring the water from the reservoir tank to the overhead tank. After that, we switch off the motor to measure the accurate readings. It is essential to wait for some time so that vibrations get damped out.

5 Results and Error Analysis

5.1 Observed readings and calculations

The main goal of the experiment was to calculate the Reynolds number. Before that, the velocity was calculated using the flow rate, and then the Reynolds number was calculated using the relation between the velocity and the Reynolds number.

5.1.1 Calculation of the velocity.

Firstly, the flow rate is calculated by keeping the volume fixed and measuring the time it took to fill the volume. The flow rate is given by

$$Q = \frac{V}{t}$$

Where,

- V is the volume occupied by the fluid(m^3).
- t is the time taken(s).

From the flow rate, velocity could be calculated using the equation below.

$$v = \frac{Q}{A}$$

- Q is the flow rate(m^3/s).
- A is the cross-sectional area of the pipe(m^2)

The diameter of the pipe was given (D=0.025 m), and from that, the cross-sectional area of the pipe was calculated, which turned out to be

$$A = \frac{\pi D^2}{4}$$

Thus,

$$A = 0.000491m^2$$

Using this data, nine different values of the velocity are achieved, which are given in the table below.

Volume(L)	Time(s)	Flow rate(ml/s)	Velocity(m/s)
0.1	12.83	7.794	0.0158
0.1	9.04	11.06	0.0225
0.1	4.92	20.32	0.0414
0.1	3.67	27.24	0.0555
0.1	3.07	32.57	0.0663
0.1	2.88	34.77	0.0707
0.1	2.79	3.58	0.0730
0.1	2.05	4.87	0.0993
0.1	1.45	6.89	0.1404

5.1.2 Calculation of the Reynolds number.

The Reynolds number depends upon the viscosity of the fluid, the density of the fluid, the diameter of the pipe, and the flow velocity. The Reynolds number is calculated using the expression shown below.

$$Re = \frac{\rho v D}{\mu}$$

Where,

- Re is the Reynolds number,
- ρ is the density of the fluid,
- v is the flow velocity,
- D is the diameter of the pipe,
- μ is the coefficient of dynamic viscosity of the fluid.

In our case, the density and the coefficient of the dynamic viscosity were taken at the temperature of 30 C and the pressure of 1 atm; the values are given below.

- $\mu = 0.000797 \text{ Pa} \cdot \text{s}$
- $\rho = 1000 \text{ Kg/m}^3$

Reynolds number can be calculated using the known value of the pipe diameter and the values shown above corresponding to the particular velocity.

Laminar flow

For the laminar flow, the Reynolds number is very low; thus, the die appears to move in

a straight path. However, because of some imperfections in the apparatus or disturbances, there may be some deviations, but they are not very significant.

Velocity (m/s)	Reynolds number
0.0158	497.95
0.0225	706.71
0.0414	1298.5

The image of the observed flow pattern in the case of laminar flow is demonstrated below.

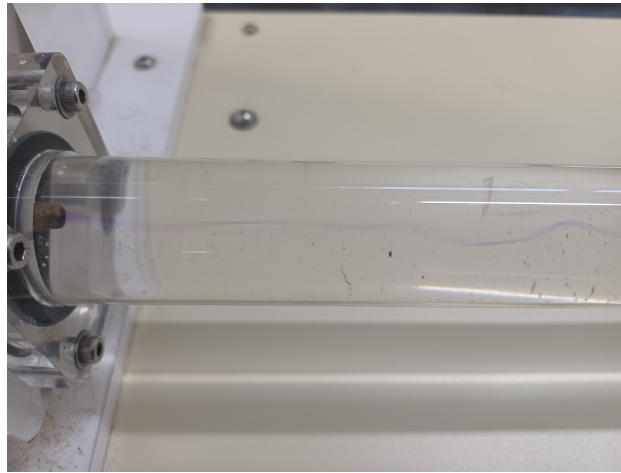


Figure 3: Laminar flow

Transition

In the case of the transition phase, the flow starts showing a deviation. It can be said that the critical Reynolds number lies in this range. However, it is difficult to predict the exact value of critical Reynolds numbers by mere observation. The values of the Reynolds number corresponding to the velocity are shown below.

Velocity (m/s)	Reynolds number
0.0555	1740.8
0.0663	2081.0
0.0707	2218.3

The image of the observed flow pattern in the case of transition is demonstrated below.

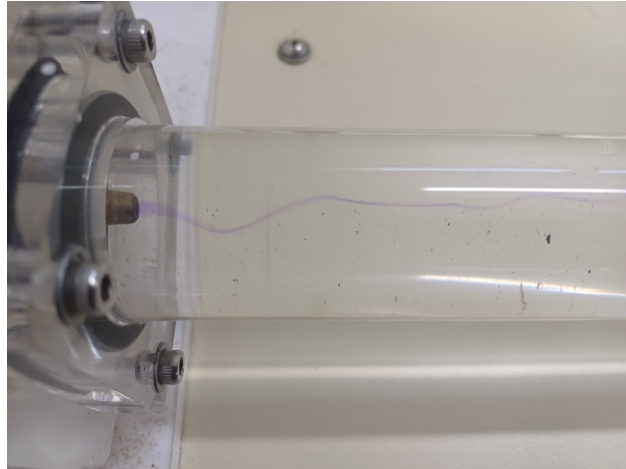


Figure 4: Transition

Turbulent flow

In the case of turbulent flow, the flow really becomes unpredictable. In this case, it is very difficult to observe any type of pattern in the flow. The calculated Reynolds number corresponding to the particular values of velocity are shown below.

Velocity (m/s)	Reynolds number
0.0730	2289.8
0.0993	3116.4
0.1404	4406.0

The image of the observed flow pattern in the case of turbulent flow is demonstrated below.



Figure 5: Turbulent flow

5.2 Error Analysis

Errors in the calculation of all the quantities are calculated using the upper bound estimate. It can be mathematically given as:

$$\sigma_Y = \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i} \right| \sigma_{x_i}$$

where,

- σ is used to indicate error,
- Y is the dependent variable,
- x_i denotes the independent variables on which Y depends.

In this case, only the errors that arise due to the instrument's least count can be calculated. However, there are other errors that are not quantifiable but contribute significantly to the deviations, but the least count errors do not have much significance in this case. Thus, least count errors may not be a good estimate to quantify the errors. Yet, the error propagation in the calculation of velocity and the Reynolds number is done for the sake of completeness.

The error propagation in velocity is given by

$$\sigma_v = \frac{V}{At^2} \sigma_t$$

Where, σ_v is the error propagated in the velocity. Similarly, the error propagation in the Reynolds number is given by

$$\sigma_{Re} = \frac{\rho D}{\mu} \sigma_v$$

Where, σ_{Re} is the error propagated in the Reynolds number. In both cases, it is assumed that the cross-sectional area of the pipe and the volume the water occupies in the measuring flask are perfect values, and they do not have any least count attached to them. The time was measured using a digital stopwatch with the least count of 0.01 s. We take $\sigma_t = 0.01s$. Thus, errors corresponding to all the readings can be given as:

Reading no.	σ_v (m/s)	σ_{Re}
1	0.0000124	0.388
2	0.0000249	0.781
3	0.0000841	2.63
4	0.000151	4.74
5	0.000216	6.77
6	0.000246	7.70
7	0.000262	8.20
8	0.000485	15.1
9	0.000969	30.3

Table 1: Error propagated due to least count of the instruments

5.3 Comparison with the theory

The generally accepted value of the critical Reynolds number is 2300 for flow in the circular pipes. In our case, the transition started happening around 1740.8, but it was not significant, and it went on increasing till the flow became completely chaotic after 2218.3. Thus, it can be said that the critical Reynolds number lies between the range of 1740.8 to 2218.3. However, it is hard to pick a particular value and say that flow becomes turbulent after this, but we can choose the fourth reading and say that the critical Reynolds number lies somewhere around that. The critical Reynolds number, in our case, is estimated to be 1740. However, it should be noted that it is just a rough estimate, and the transition happens within the range of values.

5.4 Sources of errors

1. **External Disturbances -**

The apparatus was very sensitive to the external disturbances. Thus, mild shaking could also disturb the flow, causing deviation from the theoretical value.

2. **Response error -**

When we measured the time for the specific height reached in the tank to determine the flow rate, the time was measured manually with the help of a stopwatch. So, it can lead to a delay in time measurement.

3. **Instrumental inaccuracies -**

There might be some minor defects in the experimental apparatus, such as pipes being bent and valves being loose, which could eventually cause uncertainties in the results.

4. **Parallax error -**

This error is caused when the observer takes measurements from an angle, resulting in differences from the actual reading.

5. **Imperfections in fluid -**

The fluid used to experiment was water. However, some impurities, such as dirt and air bubbles in the fluid, could contribute to the total error.

6. **Least count errors -**

The number called least count is associated with every measuring instrument; errors arising from the least count are termed as least count errors. The contribution of the least count errors to the total error depends upon the instrument's resolution.

6 Observations

The following observations were made after doing the analysis of the Reynolds Apparatus experiment.

- We cannot get a very precise value of the critical Reynolds number, but we get the range of values in which the critical Reynolds number lies. We can say that in that range, the flow transitions from the laminar to the turbulent.
- The apparatus is very sensitive to external disturbances. Thus, care should be taken while performing the experiment so that flow does not get disturbed.

7 Conclusion

We did not get the precise value of the critical Reynolds number, so we reported the estimated value and the range in which the critical Reynolds number can lie. The estimated Reynolds number 1740 is somewhat close to the theoretically expected value of 2300. However, the deviation might have arrived because of the reasons mentioned in section 5.4.

8 Acknowledgements

We want to thank our Teaching Assistant, Mr. Abhiram, for explaining the concept of Reynolds Apparatus and providing us with essential insights about the results and the sources of errors.

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